

Anshu

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SUMMARY

B.Sc. (Hons) Computer Science graduate, Hansraj College, University of Delhi (CGPA: 8.015). Strong foundation in data structures, algorithms, and system design, applied across production-style backend systems including distributed async pipelines, partitioned relational schemas, fault-tolerant API services, and CI/CD quality gates. Experienced with Docker-based infrastructure, message queues, and multi-tiered service architecture. Solved 300+ DSA problems. Led a 6-engineer team at Smart India Hackathon 2025, and independently shipped two additional hackathon projects end-to-end.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++

CS Fundamentals: Data Structures, Algorithms, OOP, System Design, Complexity Analysis

Backend & APIs: FastAPI, Node.js/Express, REST API Design, Celery, Redis (Message Queues), JWT Auth, Zod, SSE

Databases: PostgreSQL (Partitioning, pg_cron, BRIN Indexing, pgvector), MySQL, Supabase, Prisma ORM

Infra & System Practices: Docker Compose (Multi-Service Stacks), Celery/Redis (Priority Queues), Root-Cause Debugging, CI/CD Regression Gates

Tools & Practices: Docker, Git, GitHub Actions (CI/CD), Vercel, Railway, Agile, AI-Assisted Development (Claude Code)

PROJECTS

EvalOps — Distributed LLM Evaluation Infrastructure

Python, FastAPI, Celery, Redis, PostgreSQL, Docker, React, Node.js github.com/Anshu-AK-beep/evalops

- Architected an async, queue-based evaluation pipeline (FastAPI + Celery/Redis, dedicated priority queues), decoupling request handling from long-running processing for latency reduction — cut API response latency from ~5s (blocking) to near-instant; shipped as a fully containerized, one-command self-host (6-service Docker Compose stack).
- Designed a BRIN-indexed time-series schema with pg_cron-automated maintenance and composite foreign keys across partitions; built drift detection with statistical safeguards (minimum-sample-size, z-score-based) to guard against false-positive alerts at scale.
- Built a CI/CD quality gate (GitHub Actions) comparing deployments against a snapshotted baseline, with pgvector-embedding-deduplicated auto-growth of the regression test suite; shipped two client SDKs (Python, zero-dependency Node.js) validated at 87.5% agreement with human judgment.

ThinkMate — Real-Time Multi-Tiered Learning System

React, TypeScript, Node.js/Express, Supabase (PostgreSQL), JWT, SSE | Hack2Skill AI4Bharat Hackathon

github.com/Anshu-AK-beep/thinkmate

- Implemented JWT authentication with Supabase Row-Level Security for per-user data isolation, and Zod-validated API contracts across all endpoints ensuring type-safe, maintainable request handling.
- Integrated real-time SSE streaming for distributed responses; resolved a multi-layer latency issue via systematic middleware root-cause analysis, reducing response times to sub-200ms.

KrishiSaathi — Full-Stack Distributed Farming Platform

Next.js, TypeScript, PostgreSQL, Prisma ORM, Vercel, GitHub Actions

github.com/Anshu-AK-beep/KrishiSaathi

- Designed a multi-tiered REST API architecture with a Prisma ORM relational schema and Clerk-based auth, engineering a fault-tolerant data pipeline over 10,000+ soil/weather records for end-to-end request lifecycle management; deployed on Vercel with CI/CD.

EXPERIENCE & HACKATHONS

Team Leader — Smart India Hackathon (SIH) 2025

2025

Hansraj College, University of Delhi | Team of 6

- Led a 6-member cross-disciplinary team across 3 parallel modules (ML pipeline, optimization logic, REST API service) using Agile sprint planning under time-constrained conditions; solution evolved into KrishiSaathi (deployed, live).

Individual Participant — Hack2Skill AI4Bharat Hackathon

2025

Built ThinkMate — Real-Time Multi-Tiered Learning System

- Independently designed, built, and deployed a full-stack learning platform end-to-end within the hackathon's build window.

Individual Participant — India Runs Hackathon (by Redrob AI)

2025

Built Intelligent Candidate Discovery — RAG Retrieval & Reasoning Pipeline

- Independently built a 5-stage retrieval-augmented candidate ranking pipeline (FAISS, BM25, Reciprocal Rank Fusion) for India Runs' Data & AI Challenge, ranking 100,000 profiles against a job description in 28 seconds on CPU.

EDUCATION

Hansraj College, University of Delhi

2023 – 2026

B.Sc. (Hons) Computer Science | CGPA: 8.015 / 10

CERTIFICATIONS & ACTIVITIES

- Backend specialization: 300+ DSA problems solved, advanced PostgreSQL features (partitioning, BRIN, pgvector) | leetcode.com/u/Anshu-AK-beep
- HPE Software Engineering Job Simulation (Forage, Sep 2026) — REST API design & build, unit testing (Java, Spring Boot)
- Commonwealth Bank Data Science Job Simulation (Forage, Feb 2026)